

GELDIUM COLLECTIONS TRANSFORMATION

AI-Powered Collections Strategy & System Architecture

A strategic blueprint for early debt intervention, agentic AI execution, responsible governance, and measurable financial recovery.

 Prepared for: Head of Collections

 Predictive Analytics & Agentic AI

 Responsible & Ethical AI

How the System Works: Inputs & Decision Logic



1. Holistic Data Inputs

Real-time ingestion of static demographics & dynamic financial behavior.

- **Credit Health:** Revolving utilization ratio (>80% risk flag), DTI ratio.
- **Behavioral History:** 6-month repayment record, 30–59 DPD occurrences.
- **Account Metrics:** Monthly income stability, loan balance, account tenure.



2. Predictive Decision Engine

Combines statistical ML probability with dynamic business rules.

- **ML Risk Scoring:** Logistic regression models calculate delinquency likelihood.
- **Segmentation Rules:** Classifies customers into Tier 1 (Low), Tier 2 (Monitor), Tier 3 (High).
- **Path Assignment:** Determines optimal intervention channel and timing.



Fintech Data Processing & Decision Engine Concept

How the System Works: Actions & Learning Loop

1

Risk Identification

Identifies early distress flags (e.g., utilization >80% + single 30 DPD).

2

Action Execution

Triggers automated SMS reminders, credit counseling, or hardship offers.

3

Response Capture

Tracks customer engagement, payment arrangements, and promise-to-pay rates.

4

Feedback & Retraining

Model retrains on recovery outcomes to optimize future channel & offer selection.



Risk-Tiered Action Matrix

- **Tier 1 (Low Risk):** Standard billing statements; no active collection intervention.
- **Tier 2 (Moderate Risk):** Automated, friendly digital SMS/Email reminders & app nudges.
- **Tier 3 (High Risk):** Early intervention, custom restructuring offers, specialist assignment.



Continuous Learning Loop

- **Outcome Calibration:** Adjusts risk weights based on actual 30/60/90-day defaults.
- **Channel Optimization:** Learns preferred contact times and response channels per customer.
- **Fairness Auditing:** Periodic automated reviews to ensure unbiased score drift over time.

Role of Agentic AI: Autonomous vs. Human Oversight

FULLY AUTONOMOUS



Autonomous AI Agent Activities

High-volume, routine operations executed dynamically without manual intervention.

- **Continuous Monitoring:** Real-time tracking of balance spikes & utilization thresholds.
- **Automated Outreach:** Generating personalized SMS, email, and app notifications.
- **Account Prioritization:** Sorting daily collections queues based on recovery probability.
- **Standard Segmentation:** Categorizing accounts and routing to optimal digital channels.

HUMAN-IN-THE-LOOP



Human-in-the-Loop Oversight

Complex, high-sensitivity cases requiring empathy, judgment, and regulatory approval.

- **Hardship Approvals:** Reviewing and approving non-standard payment restructuring plans.
- **Complex Escalations:** Handling disputed debts, severe distress, or legal exceptions.
- **Policy & Guardrail Governance:** Override options for agents during edge cases.
- **Fairness & Strategy Audits:** Quarterly performance reviews and compliance checks.

Responsible AI Guardrails: Fairness, Explainability & Compliance



Fairness & Bias Mitigation

Preventing unintended discrimination across demographic and credit segments.

- **Demographic Parity Audits:** Testing prediction outcomes across age, location, and employment.
- **Disparate Impact Analysis:** Ensuring positive intervention offers are distributed equitably.
- **Supportive Focus:** Restricting automated actions to helpful assistance rather than punitive cuts.



Explainability & Transparency

Providing clear, human-understandable drivers behind every automated risk score.

- **SHAP Feature Attribution:** Quantifying exactly why an account was flagged for intervention.
- **Plain-Language Summaries:** Giving collections staff clear reasons (e.g., "High utilization + missed payment").
- **Customer Transparency:** Clear communication regarding available hardship programs.



Compliance & Governance

Strict alignment with financial consumer protection and privacy regulations.

- **Regulatory Adherence:** Compliant with ECOA, FCRA, GDPR, and local debt recovery laws.
- **Adverse Action Logging:** Documented audit trails for any negative credit decisions.
- **Human Appeal Path:** Right to request human review for automated hardship denials.

Expected Business Impact: Quantitative & Qualitative Outcomes

15%

DELINQUENCY REDUCTION

60%

OUTREACH AUTOMATED

40%

HARDSHIP PLAN ADOPTION

25%

LOWER RECOVERY COSTS

OUTCOME DIMENSION	TARGET METRIC / OBJECTIVE	STRATEGIC VALUE FOR GELDUM
Financial Recovery	Reduce 90+ DPD defaults by 10–15% in pilot cohort	Protects capital reserves and significantly decreases charge-off provisions.
Operational Efficiency	Automate ~60% of routine Tier 1/2 communications	Frees collections specialists to focus exclusively on complex, high-value recovery cases.
Customer Experience	Increase Net Promoter Score (NPS) in collections by +15 pts	Transforms collections into empathetic financial assistance, preserving customer lifetime value.
Governance & Risk	100% auditability & ECOA regulatory compliance	Eliminates regulatory penalties and ensures fair, transparent treatment across all borrower groups.

Image Sources



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